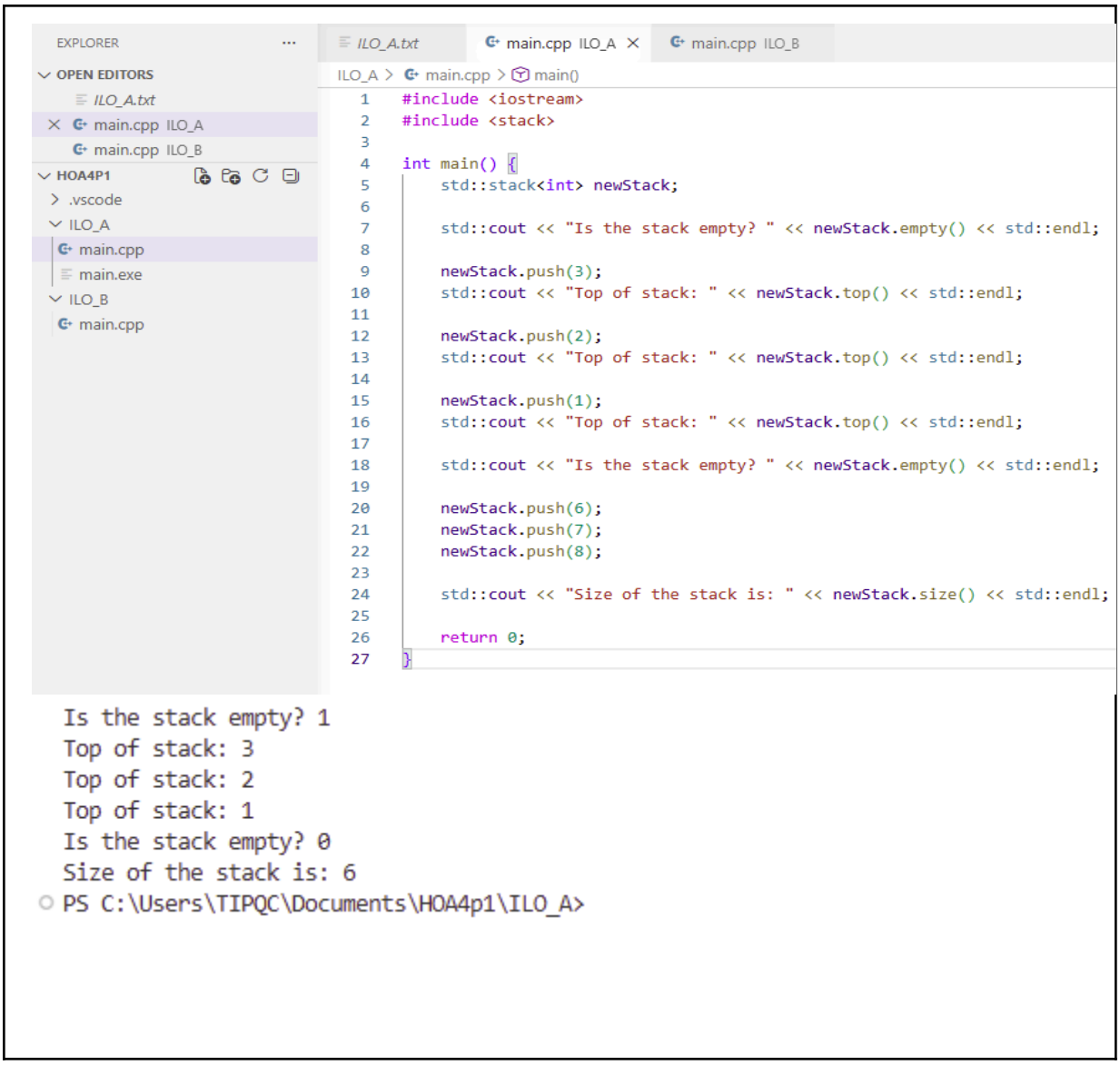
COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-CPE21S3 - Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	7/28/26	**DATE SUBMITTED:**	7/28/26
NAME:	Busa, Emmanuel A.	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. ___ [ACTIVITY NAME]

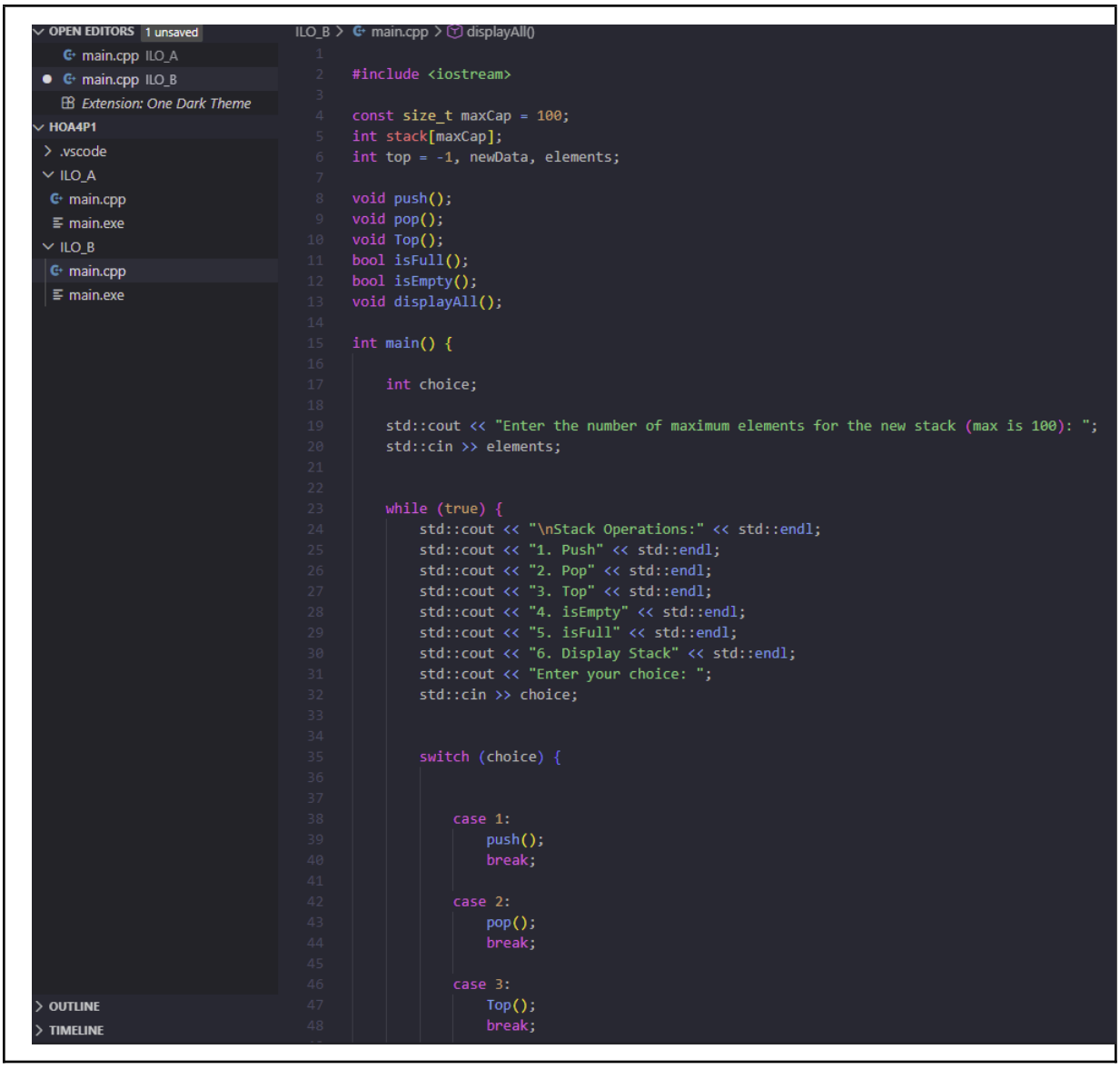
1. PROCEDURE OUTPUT

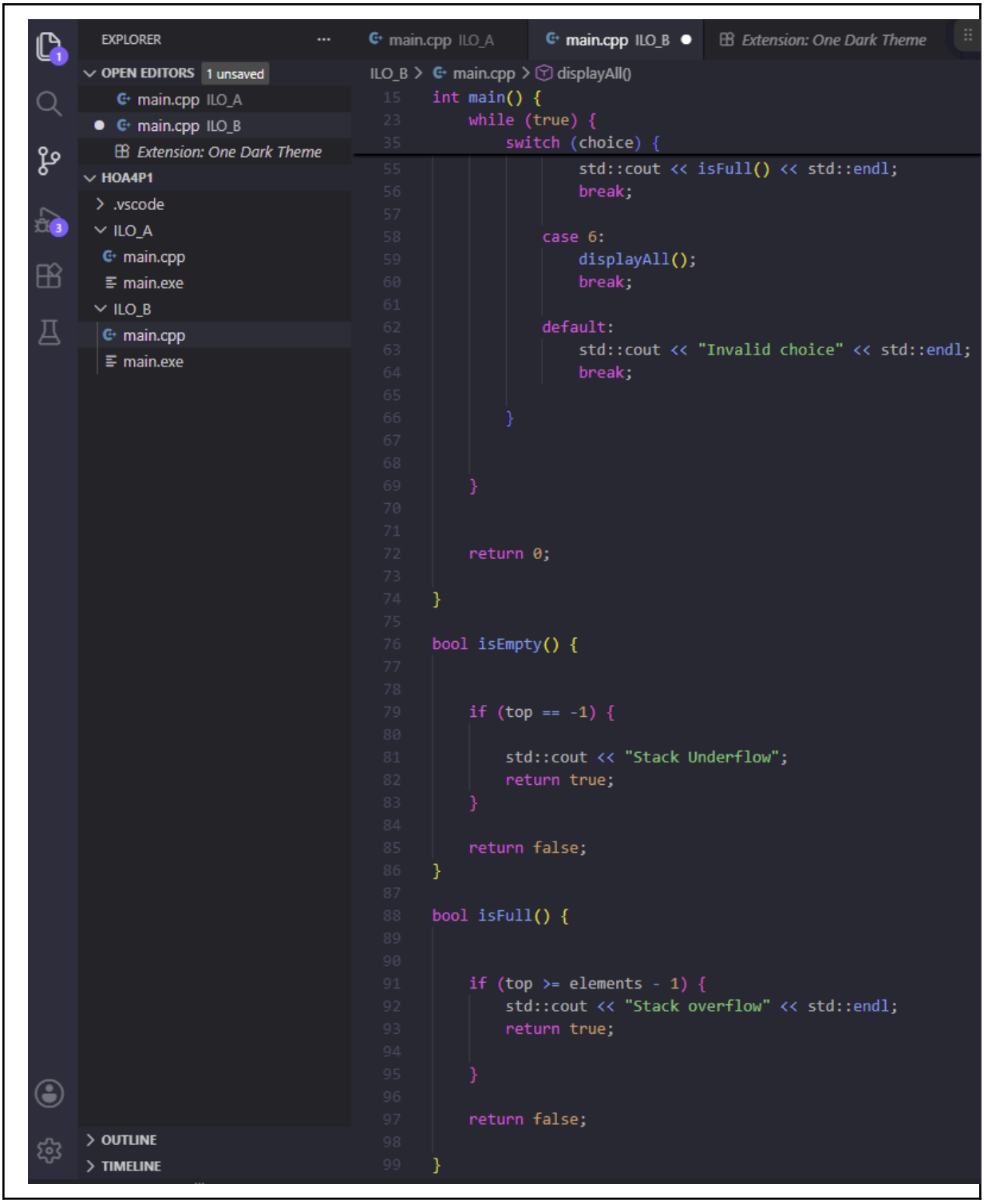


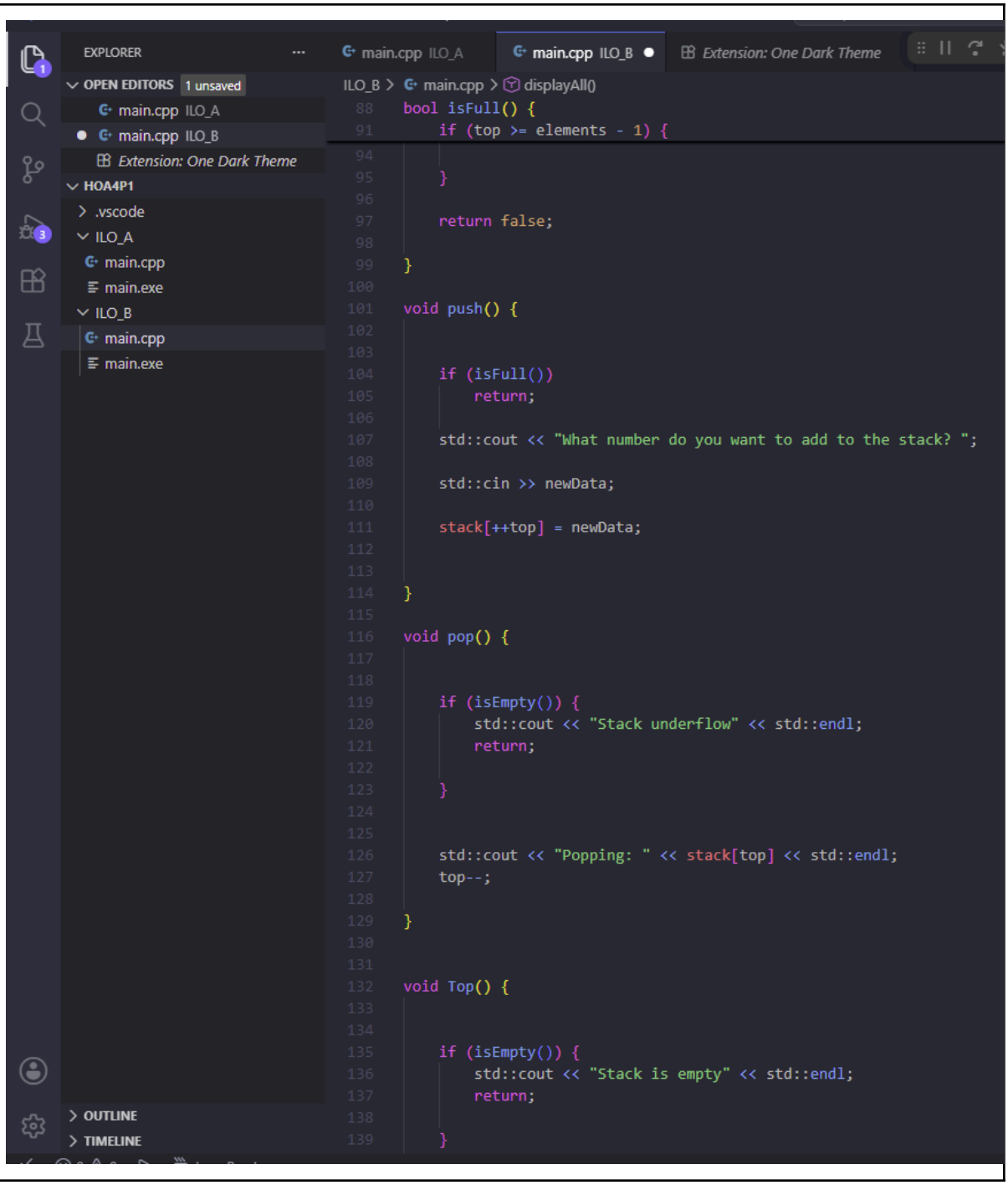
Code Analysis:

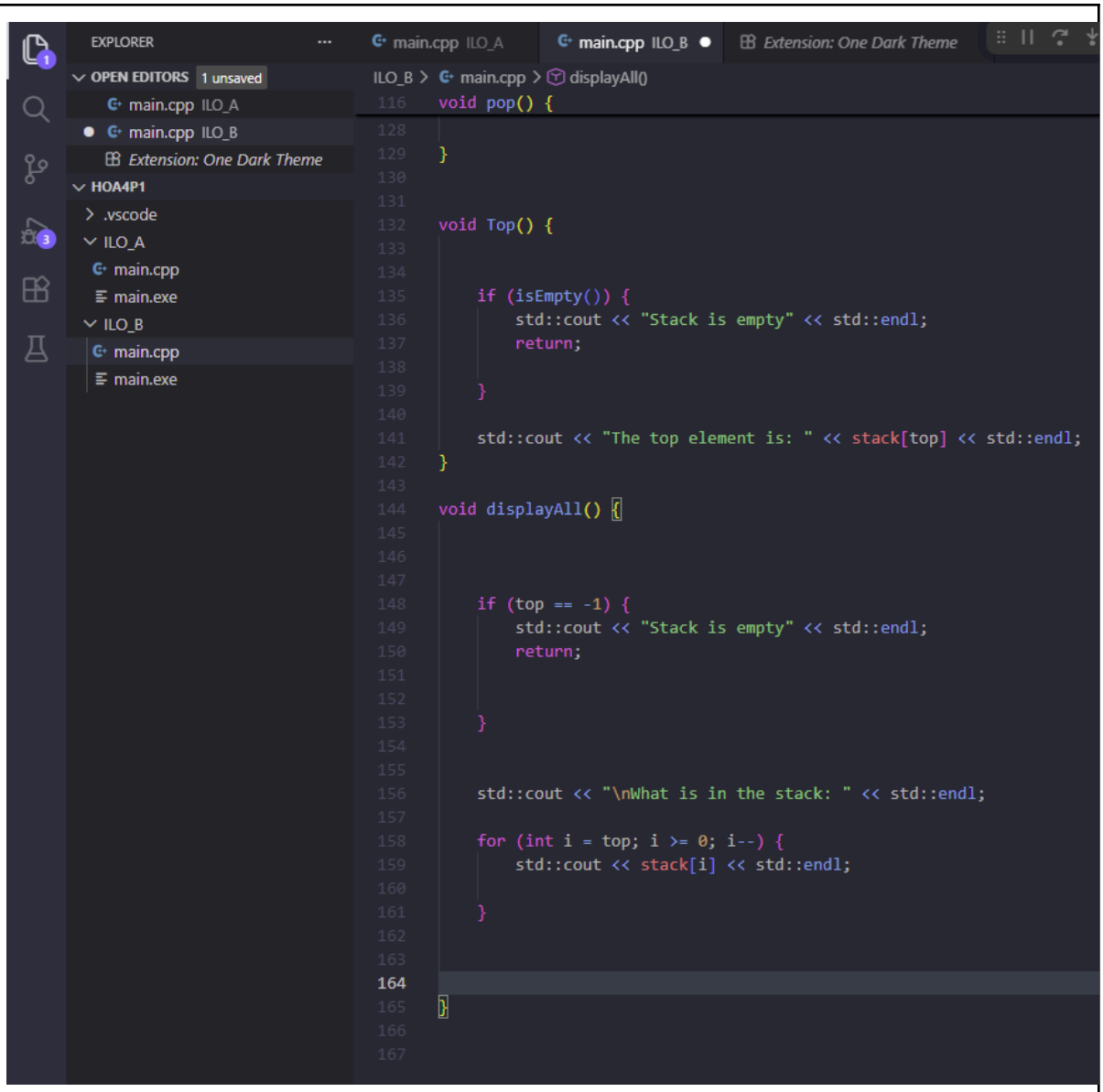
The first newStack.empty is called to check if the stack is empty and it displays 1 because it was still empty when it was first called. The newStack.top displays 3, 2 and 1 because of the cout. The second new.Stack is the next line of code to run and it displays 0 which means the code is now not empty. The newStack.size is performed to check how many elements are there in the stack and it displays 6 because of the newStack.top we put in the code.

Table 1: Output of ILO A



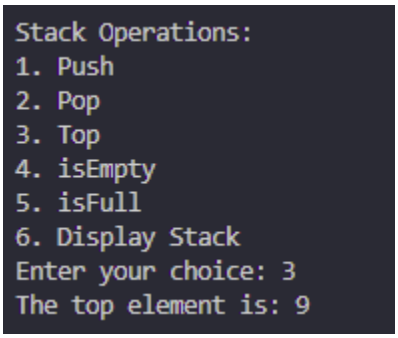

EXPLORER
main.cpp ILO_A
main.cpp ILO_B
Extension: One Dark Theme
OPEN EDITORS 1 unsaved
main.cpp ILO_A
main.cpp ILO_B
Extension: One Dark Theme
HOA4P1
.vscode
ILO_A
main.cpp
main.exe
ILO_B
main.cpp
main.exe
OUTLINE
TIMELINE
ILO_B > main.cpp > displayAll()
int main() {
while (true) {
switch (choice) {
std::cout << isFull() << std::endl;
break;
case 6:
displayAll();
break;
default:
std::cout << "Invalid choice" << std::endl;
break;
}
}
return 0;
}
bool isEmpty() {
if (top == -1) {
std::cout << "Stack Underflow";
return true;
}
return false;
}
bool isFull() {
if (top >= elements - 1) {
std::cout << "Stack overflow" << std::endl;
return true;
}
return false;
}

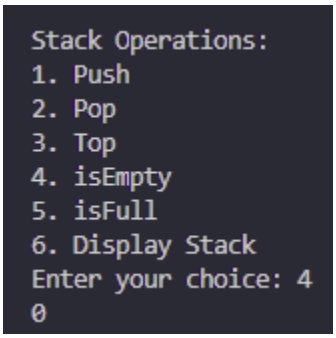

EXPLORER
OPEN EDITORS 1 unsaved
main.cpp ILO_A
main.cpp ILO_B
Extension: One Dark Theme
HOA4P1
.vscode
ILO_A
main.cpp
main.exe
ILO_B
main.cpp
main.exe
OUTLINE
TIMELINE
main.cpp ILO_A
main.cpp ILO_B
Extension: One Dark Theme
ILO_B > main.cpp > displayAll()
88 bool isFull() {
91 if (top >= elements - 1) {
94
95 }
96
97 return false;
98
99 }
100
101 void push() {
102
103
104 if (isFull())
105 return;
106
107 std::cout << "What number do you want to add to the stack? ";
108
109 std::cin >> newData;
110
111 stack[++top] = newData;
112
113
114 }
115
116 void pop() {
117
118
119 if (isEmpty()) {
120 std::cout << "Stack underflow" << std::endl;
121 return;
122
123 }
124
125
126 std::cout << "Popping: " << stack[top] << std::endl;
127 top--;
128
129 }
130
131
132 void Top() {
133
134
135 if (isEmpty()) {
136 std::cout << "Stack is empty" << std::endl;
137 return;
138
139 }

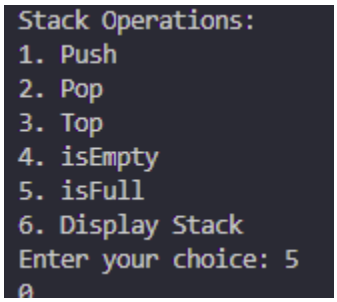


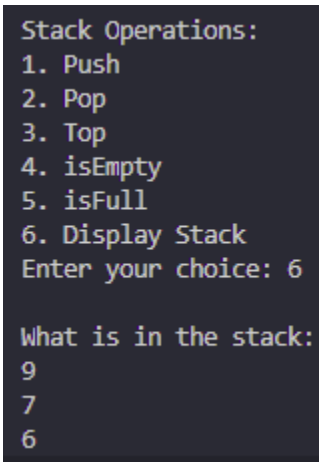

```
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display Stack
Enter your choice: 1
What number do you want to add to the stack? 67
```

```
Stack Operations:
1. Push
2. Pop
3. Top
4. isEmpty
5. isFull
6. Display Stack
Enter your choice: 2
Popping: 67
```









Code Analysis:

First, we limit our data to only be no more than 100 and we coded our variable names. The user can choose 1-6 depending on what they want to do. I coded the cases for the display. The push adds a value in our stack and it always displays the latest value the user inputted, the pop removes the top value, the top checks what is the top value, the isEmpty and isFull checks if the stack is empty or full and displays 0 for false 1 for true and displays a text if it is empty or not. The display Stack displays all of the values the user entered in the stack, it works by setting making i = top and decrementing it until it is greater than or equal to zero then it stops the loop and displays all of the values in the stack.

Table 2:Output of ILO B

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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